Clàudia Janó Muñoz

Email: claudia.jano@student.manchester.ac.uk
Mobile: +44 7935493960

Education

- **MPhys Physics with Astrophysics** — Manchester, UK
  University of Manchester — *Sep. 2018 - Present*
  - **Course performance:** 69% average in Year 1, 74.4% average in Year 2 and 77.6% average in Year 3.
  - **Relevant coursework:** Introduction to Astrophysics, Physics of the Solar System, Galaxies, Astrophysical Processes, Nuclear Fusion and Astrophysical Plasmas, Cosmology, Stars and Stellar Evolution, Exoplanets, Radio Astronomy, Gravitation, Advanced Statistical Physics
  - **Scholarships:** Physics Success Scholarship (Dec. 2018)
- **Título de Bachiller (A-level equivalent)** — Badalona, Barcelona, Spain
  Escola Garbí Pere Vergés — *2016 - 2018*
  - **Performance:** 96.5% average in Bachiller including 100% in Physics, Mathematics, Chemistry and English. 90% average in PAU (national examinations) including 100% in Physics
  - **Awards:** PAU Distinction for results in national examinations by the *Generalitat de Catalunya* (Government of Catalonia)

Experience

- **Machine learning in spectra analysis from ALMA observations** — University of Manchester, UK
  MPhys Project supervised by Prof Gary Fuller — *Sept. 2021 - Present*
  - Currently developing a machine learning model to analyse molecular spectra around protostellar regions.

- **Simulating the effects of close stellar encounters** — Universitat Autònoma de Barcelona, Spain
  Research project supervised by Dr Ignasi Ribas — *Sept. 2021 - Oct. 2021*
  - Implemented an N-body simulation of the Oort cloud based on current structure models.
  - Simulated the evolution of Oort cloud comets during past and future close encounters between the Sun and nearby stars using the REBOUND Python library.
  - Modelled the spread of intelligent life in the Milky Way using stellar encounter statistics, planet population studies and pandemic spread models.

- **Exoplanet atmospheric evolution with machine learning** — Imperial College London, UK
  UROP Summer Internship supervised by Dr James Owen — *June 2021 - Aug. 2021*
  - Developed a classical machine learning model to emulate the effects of photoevaporation on exoplanet evolution.
  - Applied algorithms such as Random Forests and Support Vector Machines to build a Super-Earth/Sub-Neptune classifier and a regressor to predict evolved exoplanet radii.
  - Inferred the initial conditions of the observed exoplanets Kepler-36b and Kepler-36c using Markov Chain Monte Carlo techniques and Bayesian statistics.
  - Gained knowledge of current planet formation and evolution models and participated in paper discussions within the Imperial Astrophysics group.
  - Generated training data with the DiRAC supercomputer and achieved 99% accuracy on emulating exoplanet population evolution, decreasing computing time by a factor of 400.
  - Presented the results to the atmospheric escape group of the University of California Santa Cruz and the ICL planets and stars group.
  - Method and results submitted for publication, to be peer-reviewed.

- **Z and Higgs boson cross section measurements** — University of Manchester, UK
  3rd Year Laboratory Experiment supervised by Prof Terry Wyatt — *March 2021 - May 2021*

- Calculated the cross section of the leptonic decays of Z and Higgs bosons using data from the ATLAS detector at CERN and obtained results within one standard deviation from literature.
- Used analysis and programming tools used in CERN such as ROOT and Monte Carlo simulations.
- Devised a user-friendly skeleton code structure to be implemented by the university across all codes for this experiment.
- Presented the results and was graded 85%.

- **Crab Pulsar timing** University of Manchester, UK

  3rd Year Laboratory Experiment supervised by Dr Patrick Weltevrede *Oct. 2020 - Nov. 2020*
  - Calculated the distance to, period, period derivative and magnetic field of the Crab Pulsar.
  - Scheduled remote observations of pulsars with the 42-ft radio telescope at the Jodrell Bank Observatory.
  - Developed code to analyse pulse timing data from known pulsars using techniques such as Fourier analysis and polynomial interpolations.
  - Gained experience presenting results in a scientific manner and was graded 75%.

- **Reconstructing the orbit of the P/2010 P4 WISE comet** Badalona, Spain

  Research Project *Sep. 2016 - Sep. 2017*
  - Measured the comet's eccentricity, period, semi-major axis and inclination from past position observations.
  - Gained experience using basic observation tools such as astronomical databases and coordinate systems.
  - Learned about celestial mechanics and Keplerian motion and its applications in the trajectory of a comet.
  - Presented in front of a jury and was graded 100%.

Skills

- **Computational skills**
  - Proficient in Python for data analysis and experienced in machine learning, Python GUIs and CERN's data analysis framework, ROOT.
  - Confident in LaTeX for scientific writing.
  - Limited experience in Windows Subsystem for Linux (WSL) and High Performance Computing usage.
- **Hardware skills**
  - Experience with general undergraduate laboratory equipment such as: circuit components, oscilloscopes and digital controller systems; lenses, light sources and other optical devices; and various other miscellaneous equipment like hot metals and heating sources.
  - Experience with some special equipment such as: optical telescopes for lunar observations; remote control of the 42ft telescope at JBCA for pulsar observations; and low radioactive sources for gamma-ray spectroscopy.
- **Languages**
  - Native speaker: Spanish and Catalan
  - Proficient: English
  - Elementary level: German (A2)

Publications

- **Submitted for peer-review**
  - *Exoplanet atmosphere evolution: emulation with random forests*, submitted to MNRAS. Accessible from arXiv: https://arxiv.org/abs/2110.15162

Extracurricular Activities

- **University of Manchester Astronomy Society** Manchester, UK

  Chair *May 2021 - Present*
  - Supervised a student committee and took responsibility in organisation and safety procedures.
  - Managed social media and event scheduling.
  - Participated in the creation of the Science Union, an outreach and social organisation aimed at natural sciences students.

  Science Officer *Feb. 2020 - May 2021*
  - Organised and hosted the Stellar Lecture Series for undergraduate students: biweekly lectures delivered by researchers about current topics in astrophysics and cosmology.
  - Gained experience hosting professionals and organisations, such as the Institute of Physics and Dame Jocelyn Bell-Burnell.
  - Planned and supervised social events such as weekly stargazing nights.